Ex No - 12b
DATE -

ECHO CLIENT-SERVER USING UDP

AIM: To implement an echo client-server application using UDP Sockets and verify communication between the client and server.

PROCEDURE:

1) Create a UDP Echo Server using DatagramSocket and DatagramPacket.
2) Assign a port number to the UDP Server.
3) Compile and execute the UDP Server program
4) Create a UDP Echo Client using Datagram Socket and Datagram packet.
5) Specify the Server address and port number.
6) Enter a message through the clients.
7) The client sends the message as a UDP datagram to the server
8) The Server receives the datagram and sends the echoed message back.
9) Display the received echoed message at the client.
10) Enter bye to terminate the communication.

1) Echo Client - Server UDP:

```
import java.net.*;
public class UDPEchoServer{
public Static void main(String[] args) throws
        Exception{
DatagramSocket Server = new DatagramSocket
                        (5001);
byte[] Buffer = new byte[1024];
System.out.println("UDP Server waiting...");
while(true){DatagramPacket request = new
            DatagramPacket(Buffer, Buffer.
  length); Server.receive(request);
  String msg = new String(request.getData()
  0, request.getLength());
if(msg.equalsIgnoreCase("bye")) break;
DatagramPacket reply = new DatagramPacket(
request.getData(), request.getLength(),
  request.getAddress(), request.getPort());
Server.send(reply); } Server.close(); } }
```

OUTPUT: UDP server waiting....

2) UDP Echo Client:

```
import java.net.*;
import java.util.Scanner;
public class UDPEchoClient{
public static void main(String[] args)throws Exception
{ DatagramSocket Client = new DatagramSocket();
InetAddress address = InetAddress.getByName("
                                localhost");
  Scanner sc = new Scanner(System.in);
while (true){ System.out.print("Enter message:");
  String msg = sc.nextLine();
  byte[] data = msg.getBytes();
DatagramPacket request = new DatagramPacket(
      data, data.Length, address, 5001);
  Client.send(request);
if(msg.equalsIgnoreCase("bye")) Break;
  Byte[] buffer = new byte[1024];
DatagramPacket = new DatagramPacket(
    data, data.length, address, 5001);
    Client.send(request);
System.out.println("Echo "+new String(reply.
          getData(), 0, reply.getLength())); }
  Client.close(); sc.close(); } }
```

OUTPUT:
```
Enter message : Hello
Echo : Hello
Enter message: Welcome
Echo : Welcome
Enter message : bye
```

RESULT: Thus, the UDP Echo client-Server application was successfully implemented and the client successfully received the echoed message from the server.